

Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	All Python and HTML files use consistent 4-space indentation throughout
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into dataset, model, templates folders with separate files for training and app
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like X_train, y_test, flood_model, input_data clearly reflect their purpose
4	Function / Method Names	Functions are descriptively named	Yes	Functions named home(), predict(), about() clearly describe their purpose
5	Code Comments	Inline and block comments explain log c	Yes	All major sections in train_model.py and app.py are commented with step descriptions
6	Modular Design	Code is split into reusable functions/modules	Yes	Training code is separate in train_model.py and web app is separate in app.py

7	No Redundant Code	Duplicate or unused code is removed	Yes	Each function performs a single specific task with no duplicate logic
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Flask handles invalid routes and form validation is done via required fields in HTML

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	flood_model.pkl	Python, XGBoost	Loaded in app.py for every prediction	High
2	Predict() function	Python, Flask	Called every time user submits the form	High
3	index.html form	HTML, CSS	Base template reusable for similar prediction projects	High
4	train_model.py	Python, Scikit-learn	Can be reused to retrain model with new data anytime	High
5	result.html	HTML, CSS, Jinja2	Reusable result page for any binary classification output	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean folder structure with separate files for training, app and templates
Readability	5	Meaningful variable names, proper indentation and section comments throughout
Reusability	4	Core modules like model, predict function and templates are highly reusable
Documentation / Comments	4	All major steps are commented clearly in both train_model.py and app.py
Overall Score	4.5	Well structured, readable and maintainable codebase suitable for production use